Tasks

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

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Network Threat	Real-World Example
Phishing	A fake Paytm login page steals your username and password.
DDoS Attack	Instagram becomes slow because attackers flood its servers with traffic.
Man-in-the-Middle (MITM)	An attacker intercepts data when using public Wi-Fi to access WhatsApp.
Malware	Downloading a fake app installs malware that steals personal data.
Password Attack	Someone repeatedly tries to guess your Instagram password.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct PIN.

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```
correct_pin = "1234" attempts = 0

for i in range(10000):     guess = f"{i:04d}"     attempts += 1

    if guess == correct_pin:
```

```
print("PIN Found:", guess)        print("Attempts:", attempts)
break
```

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

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Risk	Mitigation
Unauthorized entry	Install biometric or key-card access.
Fire	Install smoke detectors and fire extinguishers.
Theft of equipment	Install CCTV cameras and lock server racks.
Power failure	Use a UPS and backup generator.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

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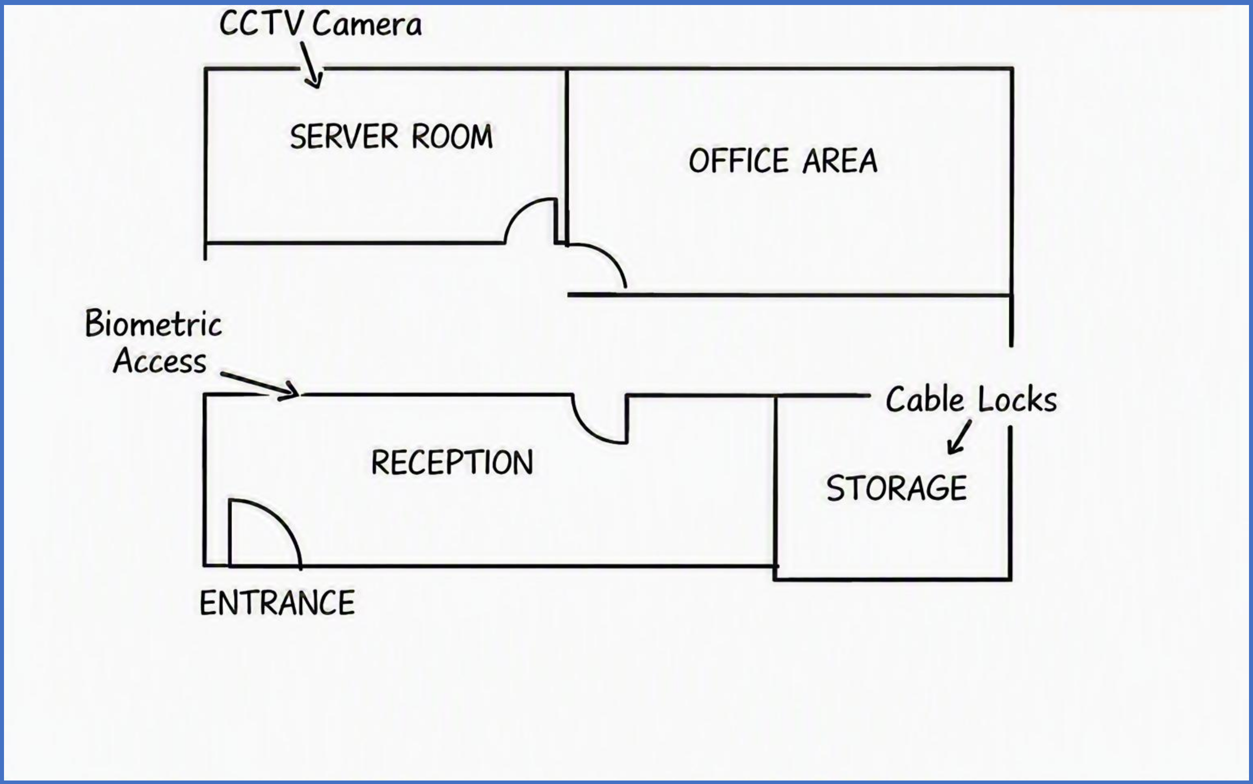

5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment

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Use strong passwords for all computers and network devices.

Install and regularly update antivirus software.

Enable Windows Firewall on every computer.

Allow only authorized students and staff to access the lab.

Lock the server room with biometric or key-card access.

Install CCTV cameras to monitor the computer lab.

Regularly back up important files and keep operating systems and software updated.

Edited for a College Lab:
This checklist is specifically designed for a college computer lab where multiple students use shared computers. It helps protect both the network (firewalls, antivirus, updates) and the physical equipment (CCTV, locked server room, controlled access).